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PI Name:	Bailey, Susan M. Ph.D.		
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PI Email:	susan.bailey@colostate.edu	**Fax:**	FY 970-491-7742
PI Organization Type:	UNIVERSITY	**Phone:**	970-491-2944
Organization Name:	Colorado State University		
PI Address 1:	Environmental and Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
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Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
Contact Email:	jbcharle@gmail.com		
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Task Description:

The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence, which together with associated proteins serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks--DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability.
It is well established that telomere length diminishes with cellular division and normal aging. It is also becoming increasingly appreciated that telomere length is influenced by a variety of other factors including life stress, infection, inflammation, and oxidative stress. Nobel Laureate Elizabeth Blackburn suggests that telomere shortening “powerfully quantifies life’s insults.” Indeed, recent research supports telomere length maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. Thus, the rate at which telomeres shorten provides an informative biomarker of biological aging and further, erosion of telomere length can be linked to age-related pathologies including reduced immune function, loss of fertility, cardiovascular disease, and cancer.

Healthy lifestyles and diets are positively correlated with telomere length, and changes in diet and lifestyle have also been shown to modulate telomerase activity. Telomerase is the specialized enzyme present in germ line, stem, and cancer cells, capable of de novo addition of telomeric repeats onto the ends of newly synthesized telomeres. A large remaining puzzle in human populations is the relationship between telomerase activity (e.g., measured in peripheral blood lymphocytes) and lifestyle factors (e.g., nutritional, psychological, and physical stressors). We speculate that for the astronauts, telomere length represents a particularly relevant biomarker, because it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut’s genetic susceptibilities, exposures to galactic cosmic radiation, physical, and psychological stressors encountered, are all “captured” as changes in telomere length.

A remarkable once-in-a-life time opportunity currently exits to evaluate the effects of spaceflight factors on the monozygotic twin astronauts Scott and Mark Kelly, who will have spent a total of 540 and 54 days in low Earth orbit, respectively. We propose to assess differential effects on telomere length and telomerase activity associated with the upcoming yearlong International Space Station (ISS) mission in the twins – individuals matched genetically and who share similar, unique lifestyles; i.e., they are of similar “nature and nurture.” Standard assays will be employed to determine telomere length and telomerase activity as part of their respective and concurrent biodosimetry/chromosome analyses pre-flight (to establish baselines), during flight (to evaluate short-term/temporary changes), and post-flight (to evaluate long-term/permanent changes). Results from the space- and ground-based twins will be compared to evaluate effects of spaceflight factors on these informative biomarkers. Results from the twins will also be compared with our current retrospective analyses of a cohort of unrelated astronauts who have participated in missions of 3 to 6 months duration to evaluate inter-individual sensitivity to these effects of spaceflight, a strategy that will also facilitate evaluation of time dependent responses.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Introduction:
This study is part of an integrated compilation of multiple studies led by multiple Principal Investigators at multiple centers that takes advantage of a remarkable and unique opportunity to investigate the effects of space travel with monozygotic twin astronauts, who will have spent a total of 540 and 54 days in low Earth orbit; importantly, they are of similar “nature and nurture.” For the purposes of the Twins study, the flight subject will be on-orbit (International Space Station--ISS) for one year, while the ground subject will remain Earth-bound for that same year. This particular project will integrate changes in telomere length and telomerase activity associated with prolonged and combined exposures to galactic space radiations, microgravity, and other spaceflight stressors (e.g., nutritional, psychological, physical). Samples from each of the twins will be assessed for telomere length and telomerase activity at similar times before the flight, during flight, and after return to Earth, to determine whether observed changes differ relative to each other (i.e., are spaceflight specific), as well as how they relate to physiological, molecular, genetic, and biochemical changes (information obtained via data sharing with other twins studies).

Telomere length maintenance captures the interplay between genetics and life experiences/exposures, and so is a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and age related degenerative diseases, ranging from reduced immune function, to cardiovascular disease and cancer. Therefore, we propose that for the astronauts, telomere length maintenance represents a particularly relevant biomarker, as it reflects the combined exposures and experiences encountered during space travel.

Our overall goal is to test the hypothesis that accelerated telomere shortening and elevated telomerase activity will be associated with spaceflight.

Specific Aims

Aim 1. Assess differential effects on telomere length and telomerase activity concurrently in the space- and ground-based twin astronauts, and in ground-based age-matched controls.

• Blood samples will be taken pre-flight (to establish baseline), during flight (to evaluate short-term/temporary changes), and post-flight (to evaluate long-term/permanent changes).

• Comparisons between the twins controls variables of individual genetic differences, susceptibilities and lifestyle factors as much as possible, making observed differences spaceflight specific.

• Comparisons with and between unrelated astronauts and age-matched controls will allow evaluating role of genetics and individual susceptibilities (on-going prospective & retrospective study of a cohort of unrelated astronauts who have participated in missions of 3-6 months duration, which includes the flight subjects' previous flight).

• Coordinated with previous and on-going biodosimetry studies; chromosome aberration analyses.

Aim 2. Initiate in vitro studies to investigate potential mechanisms (e.g., oxidative stress) and mitigation strategies (e.g., antioxidants).

Materials and Methods:

	Blood samples will be taken pre-flight (to establish baseline), in-flight (to evaluate short-term/temporary changes), and post-flight (to evaluate long-term/permanent changes). Telomere length is being assessed using TELO-FISH; Florescence in situ Hybridization (FISH) with a telomere probe on chromosomes (and interphase nuclei) is evaluated as Relative Fluorescence Intensity (RFI) and reveals distributions of telomere length (shortest to longest). Telomerase activity is being assessed using the quantitative Real Time-PCR Telomere Repeat Amplification Protocol (qRT-PCR TRAP). For telomere length measurements, for each individual at each sampling time point (pre-flight, in-flight, post-flight), a minimum of 50 cells will be evaluated to establish baselines and determine significant changes. A number of different comparisons will be considered: 1) between the twins to evaluate effects of spaceflight on telomere length and telomerase activity; and 2) between the twins and unrelated astronauts and controls to evaluate inter-individual sensitivity and time dependent responses to these effects of spaceflight. We anticipate significant telomere shortening and elevated telomerase activity to be associated with increased exposure to spaceflight factors (other factors being equal in the twins). We also anticipate a strong genetic component, in that certain individual astronauts will be more susceptible to telomere shortening and/or telomerase activation than others; if proven true, it would then be possible to make informed associations between particularly susceptible individuals and their specific exposures and/or medical histories. Individual chromosome aberration analyses will also be informative in this regard. Investigation of potential mechanisms (e.g., oxidative stress) and intervention strategies (e.g., antioxidants) will be initiated with in vitro experimentation combining various treatment strategies and assessment of telomere length and telomerase activity (as appropriate). General approach will be: 1) induction of oxidative stress (e.g., Hydrogen Peroxide) and alleviation of oxidative stress (antioxidant treatment; e.g., Resveratrol, Protandim); 2) evaluation of senescence and stem cell fractions associated with changes in telomere length and telomerase activity, respectively. Results: Pre-Flight samples on both the inflight and ground subject (and ground-based age-matched control subjects) are currently being collected and analyzed. Discussion: We are currently processing samples to run our analysis. Data sharing for other endpoints and coordinating with other investigators will also inform this effort.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Abstracts for Journals and Proceedings	Bailey SM, Sishc BJ, Nelson C, McKenna M, Taylor L, George K. "Assessing Telomere Length and Telomerase Activity in Astronauts." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. Abstract #0142. , Jan-2015